Customer Shopping Behavior Analysis

Project Objective

The objective of this project is to analyze consumer shopping behavior using transactional data from 3,900 purchases to identify trends in customer spending, product preferences, customer segments, and subscription patterns. The project aims to understand how factors such as discounts, customer reviews, seasons, payment methods, and sales channels influence purchasing decisions and repeat purchases, enabling the company to improve customer engagement, optimize marketing strategies, and make data-driven business decisions.

About the dataset

- Rows: 3,900

- Columns: 18

- Key Features: - Customer demographics (Age, Gender, Location, Subscription Status) - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color) - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type) - Missing Data: 37 values in Review Rating column

Exploratory Data Analysis using Python

Python was used to explore and prepare the dataset for analysis.

- **Data Loading and Initial Exploration**: Imported the dataset using Pandas and examined its structure, data types, and statistical summaries using functions such as df.info() and df.describe().
- **Missing Value Treatment**: Identified null values and filled missing entries in the review_rating column using the median rating for each product category.
- **Column Renaming**: Converted column names to snake_case to improve readability and maintain consistency.
- **Feature Engineering:**
    - Created an age_group column by categorizing customer ages.
    - Created a purchase_frequency_days column using purchase data.
- **Data Consistency** Check: Checked for redundant information between discount_applied and promo_code_used and removed the promo_code_used column.
- **Database Integration**: Connected Python to MySQL and imported the cleaned dataset into the database for further SQL analysis.

Data Analysis using SQL

We performed structured analysis in MySQL to answer key business questions:

1. **Gender-Based Revenue Split**: How does total revenue differ between male and female customers?

gender	revenue
Male	157890
Female	75191

2. **Big Spenders Using Discounts**: Which customers applied a discount to their order and still spent more than the average purchase amount?

customer_id	purchase_amount
1662	86
1667	64
1671	73
1673	73
1674	62
1676	90

839 row(s) returned

3. **Top 5 Products by Rating** – Found products with the highest average review ratings.

item_purchased	Average Product rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Handbag	3.78

4. **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

shipping_type	round(AVG(purchase_amount),2)
Express	60.48
Standard	58.46

5. **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

subscription_status	total_customers	avg_spend	total_revenue
Yes	1053	59.49	62645
No	2847	59.87	170436

6. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

item_purchased	discount_rate
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

7. **Top 3 Products per Category** – Listed the most purchased products within each category.

item_rank	category	item_purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169

8. **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

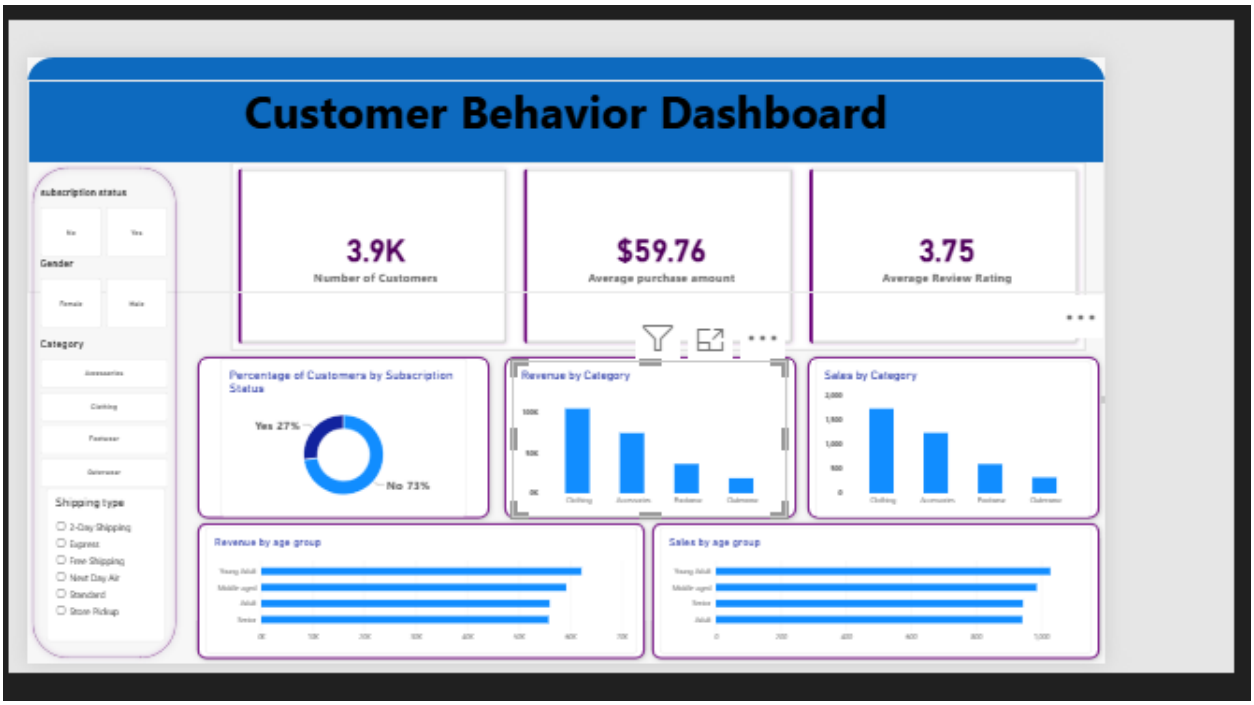
subscription_status	repeat_buyers
Yes	958
No	2518

9. **Revenue by Age Group** – Calculated total revenue contribution of each age group.

age_group	total_revenue
Young Adult	62143
Middle-aged	59197
Adult	55978
Senior	55763

Dashboard in Power BI

Finally, we built an interactive dashboard in Power BI to present insights visually.



Business Recommendations

- **Grow the Subscriber Base**: Offer exclusive perks and benefits to encourage more customers to subscribe.
- **Reward Repeat Customers**: Introduce loyalty programs that encourage repeat buyers to progress into the “Loyal” segment.
- **Rethink Discounting Strategy**: Reassess discount policies to keep sales growing without eroding profit margins.
- **Spotlight Winning Products**: Feature top-rated and best-selling products prominently in marketing campaigns.
- **Focus on High-Value Audiences**: Direct marketing efforts toward the age groups that bring in the most revenue and customers who prefer express shipping.